

Model Lab

Every prior, same slate — 2026-09-14

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

35,816

Held-Out Scores

Fitted on plate appearances before 2026-08-09, scored on the 35,816 that came after (actual HR rate 2.99%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5657	1.32×	0.02894	0.13360	0.00482
KNN comps Prior is the pooled rate of the nearest comparables.	0.5844	1.52×	0.02894	0.13343	0.00548
SVR Support-vector regression on the same feature matrix.	0.5739	1.35×	0.02905	0.13554	0.00922
Random forest Forest regression; a learned similarity metric.	0.5764	1.35×	0.02900	0.13462	0.00820
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5844	1.52×	0.02894	0.13343	0.00548
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5832	1.46×	0.02895	0.13357	0.00571
Linear regression Ordinary least squares on the same features.	0.5813	1.35×	0.02898	0.13425	0.00717
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5831	1.47×	0.02898	0.13384	0.00736
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5821	1.35×	0.02897	0.13378	0.00768
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5750	1.37×	0.02906	0.13509	0.00942

Blend weights (forest variant): KNN 0.80, SVR 0.00, RF 0.20

Shrinkage K per prior: league 250, knn 250, svr 250, rf 250, core 250, ensemble 250, linear 250, logistic 250, nn 250, xgb 250 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,789 PA, actual HR rate 2.45%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5094	0.64×	0.02398	0.11588	0.00881
KNN comps Prior is the pooled rate of the nearest comparables.	0.5673	1.07×	0.02409	0.11763	0.01438
SVR Support-vector regression on the same feature matrix.	0.5475	0.86×	0.02479	0.13057	0.02953
Random forest Forest regression; a learned similarity metric.	0.5441	0.75×	0.02446	0.12504	0.02356
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5673	1.07×	0.02409	0.11763	0.01438
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5629	0.97×	0.02414	0.11854	0.01554
Linear regression Ordinary least squares on the same features.	0.5617	0.86×	0.02439	0.12425	0.02196
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5657	0.86×	0.02429	0.12006	0.02145
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5663	0.75×	0.02424	0.11946	0.01998
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5493	0.75×	0.02498	0.12841	0.03279

Slate Agreement

Flat league · 3/6 shared 2/6 hit

Jackson Merrill

Pete Crow-Armstrong ✓

Alex Bregman

Fernando Tatis Jr. ✓

Adael Amador

Matt Olson

SVR · 4/6 shared 1/6 hit

Jackson Merrill

Pete Crow-Armstrong ✓

Matt Olson

Adael Amador

Mickey Moniak

Griffin Conine

Core (KNN+SVR) · 6/6 shared 1/6 hit

Jackson Merrill

Pete Crow-Armstrong ✓

Matt Olson

Kyle Stowers

Mickey Moniak

Heriberto Hernández

Linear regression · 5/6 shared 1/6 hit

Emmanuel Rodriguez

Jackson Merrill

Pete Crow-Armstrong ✓

Matt Olson

Heriberto Hernández

Mickey Moniak

Neural net (torch) · 5/6 shared 1/6 hit

KNN comps · 6/6 shared 1/6 hit

17.4% Jackson Merrill

15.2% Pete Crow-Armstrong ✓

15.1% Matt Olson

15.0% Kyle Stowers

14.8% Mickey Moniak

14.7% Heriberto Hernández

Random forest · 4/6 shared 1/6 hit

20.6% Jackson Merrill

18.6% Emmanuel Rodriguez

18.4% Pete Crow-Armstrong ✓

18.1% Matt Olson

17.8% Josue De Paula

17.5% Mickey Moniak

Forest blend · 5/6 shared 1/6 hit

20.5% Jackson Merrill

17.4% Pete Crow-Armstrong ✓

17.1% Matt Olson

16.5% Emmanuel Rodriguez

16.3% Kyle Stowers

16.3% Mickey Moniak

Logistic regression · 4/6 shared 2/6 hit

25.1% Emmanuel Rodriguez

20.4% Jackson Merrill

17.2% Pete Crow-Armstrong ✓

17.1% Matt Olson

16.4% Aaron Judge ✓

16.3% Mickey Moniak

20.5%

17.4%

17.1%

16.5%

16.3%

16.3%

20.8%

18.6%

17.8%

17.7%

17.3%

16.8%

20.5%

17.5%

17.2%

16.5%

16.4%

16.4%

25.4%

21.5%

18.6%

18.4%

17.4%

17.2%

Heriberto Hernández

16.7%

Mickey Moniak

17.1%

Hitter by Hitter

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)	XGBoost
1. Jackson Merrill slate San Diego Padres · vs RHP · 25 HR / 614 PA	17.4%	20.5%	20.6%	20.8%	20.5%	20.5%	20.4%	21.5%	20.7%	20.6%
2. Pete Crow-Armstrong slate Chicago Cubs · vs LHP · 41 HR / 672 PA	15.2%	17.4%	18.6%	17.8%	17.4%	17.5%	17.2%	18.6%	17.8%	17.7%
3. Matt Olson Atlanta Braves · vs RHP · 39 HR / 650 PA	14.7%	17.1%	18.4%	17.7%	17.1%	17.2%	17.1%	18.4%	17.8%	17.7%
4. Kyle Stowers slate Miami Marlins · vs RHP · 19 HR / 472 PA	13.5%	16.5%	16.7%	16.0%	16.5%	16.4%	15.3%	16.4%	16.3%	16.2%
5. Mickey Moniak Colorado Rockies · vs RHP · 22 HR / 403 PA	14.2%	16.3%	17.8%	16.8%	16.3%	16.4%	16.3%	17.2%	17.0%	16.9%
6. Heriberto Hernández Miami Marlins · vs RHP · 24 HR / 450 PA	14.0%	16.3%	16.5%	16.5%	16.3%	16.3%	16.4%	17.0%	16.7%	16.6%
7. Emmanuel Rodríguez Minnesota Twins · vs RHP · 1 HR / 8 PA	10.3%	15.9%	9.1%	18.6%	15.9%	16.5%	25.1%	25.4%	16.8%	16.7%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)	XGBoost
8. Griffin Conine Miami Marlins · vs RHP · 15 HR / 280 PA	13.1%	15.9%	17.5%	16.2%	15.9%	15.9%	16.2%	16.7%	16.7%	16.7%
9. Ronald Acuña Jr. Atlanta Braves · vs RHP · 17 HR / 426 PA	14.5%	15.9%	15.8%	15.6%	15.9%	15.8%	15.5%	15.6%	15.8%	15.8%
10. Hunter Goodman Colorado Rockies · vs RHP · 38 HR / 531 PA	14.4%	15.7%	16.6%	16.5%	15.7%	15.9%	15.7%	16.3%	16.4%	16.4%
11. Riley Greene Detroit Tigers · vs RHP · 20 HR / 540 PA	13.2%	15.6%	15.5%	15.5%	15.6%	15.5%	14.9%	15.6%	15.5%	15.5%
12. Pete Alonso slate Baltimore Orioles · vs RHP · 35 HR / 652 PA	13.3%	15.5%	15.9%	15.6%	15.5%	15.6%	15.3%	16.0%	15.5%	15.5%
13. Fernando Tatis Jr. San Diego Padres · vs RHP · 21 HR / 649 PA	15.0%	15.5%	14.6%	14.8%	15.5%	15.4%	15.8%	16.1%	15.8%	15.8%
14. Manny Machado San Diego Padres · vs RHP · 27 HR / 621 PA	14.2%	15.5%	16.1%	15.8%	15.5%	15.5%	15.6%	15.7%	15.8%	15.8%
15. Juan Soto slate New York Mets · vs RHP · 25 HR / 421 PA	12.4%	15.4%	15.9%	15.6%	15.4%	15.4%	15.2%	15.6%	15.5%	15.5%
16. Aaron Judge slate New York Yankees · vs RHP · 17 HR / 279 PA	11.3%	15.3%	16.6%	15.9%	15.3%	15.4%	16.2%	17.4%	16.3%	16.3%
17. Seiya Suzuki Chicago Cubs · vs LHP · 24 HR / 583 PA	14.0%	15.3%	15.5%	15.3%	15.3%	15.3%	14.9%	15.3%	15.1%	15.1%
18. Josue De Paula Los Angeles Dodgers · vs LHP · 1 HR / 11 PA	10.7%	15.2%	10.1%	17.3%	15.2%	15.6%	14.6%	11.0%	13.6%	13.6%
19. Kazuma Okamoto Toronto Blue Jays · vs RHP · 32 HR / 588 PA	13.1%	15.0%	16.0%	15.2%	15.0%	15.0%	14.8%	15.6%	15.4%	15.4%
20. Max Muncy Los Angeles Dodgers · vs LHP · 28 HR / 517 PA	12.2%	14.9%	15.9%	15.5%	14.9%	15.0%	15.3%	16.7%	15.7%	15.7%
21. Alex Bregman Chicago Cubs · vs LHP · 25 HR / 669 PA	15.1%	14.7%	15.0%	15.1%	14.7%	14.7%	15.4%	15.4%	15.1%	15.1%
22. Ty France	13.5%	14.6%	14.0%	14.1%	14.6%	14.5%	14.4%	14.2%	14.2%	14.2%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)	X
37. Miguel Vargas Chicago White Sox · vs LHP · 30 HR / 648 PA	11.2%	13.7%	13.5%	13.1%	13.7%	13.6%	13.7%	14.3%	13.9%	
38. Teoscar Hernández Los Angeles Dodgers · vs LHP · 13 HR / 437 PA	11.6%	13.7%	13.3%	13.5%	13.7%	13.6%	13.1%	13.2%	13.2%	
39. Spencer Torkelson Detroit Tigers · vs RHP · 23 HR / 562 PA	11.7%	13.6%	13.1%	12.9%	13.6%	13.5%	12.9%	13.1%	13.1%	
40. Brett Callahan Detroit Tigers · vs RHP · 1 HR / 53 PA	11.2%	13.4%	8.4%	10.3%	13.4%	12.8%	10.6%	10.2%	11.1%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.